Gitesh Kumar

Applied AI and Systems Engineer

Systems & Development Engineer with a dual background in Mechanical Engineering and Applied AI. I offer 4+ years of industrial experience in safety-critical environments (Oil & Gas) specializing in the integration of high-frequency data from SCADA and ERP systems. I like building reliable data pipelines and validating complex models for real-world deployment using Safety-by-design principles.

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EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg — From October 2022 to March 2026
Masters in Elektromobilität-ACES
Specialization: AI and Sustainable Production Technology

PEC University of Technology Chandigarh, India — From July 2013 to May 2017
Bachelors of Engineering
Mechanical Engineering

WORK EXPERIENCE

Sr. Operations Officer- Data and Analytics — From July 2017 to August 2021 **(4 years+)**
Hindustan Petroleum Corporation Limited India

- Managed the end-to-end integration of high-frequency data from SCADA readings and ERP (JD Edwards) systems by acting as the primary for recording, storing and preparing industrial data for operational processing.
- Served as the technical coordinator for field-testing and operational validation of industrial vision systems in collaboration with Honeywell.
- Replaced error-prone manual reporting with structured digital data-flow using Python ensuring that critical production data was processed without latency and errors for daily decision support.
- Utilized data-driven monitoring to identify bottlenecks, contributing to a 10% improvement in production efficiency while maintaining a Zero-Failure safety protocol.

RECENT ACADEMIC PROJECTS

Mini Thesis: Explainable AI in Computer Vision for End-of-Line Inspection (FAPS Lab)

- Designed and implemented a computer vision based defect detection system for end-of-line inspection using YOLOv8, targeting missing, loose and incorrectly assembled components in industrial assemblies.
- Built a full data pipeline from image acquisition to evaluation following CRISP-DM / DMME methodologies for engineering applications.
- Integrated multiple Explainable AI (XAI) techniques like Grad-CAM, Guided Grad-CAM, LIME, Layer-wise Relevance Propagation (LRP) and Eigen-CAM to analyze and visualize model decision logic at inference time.
- Demonstrated how explainability improves trust, debugging and acceptance of vision systems in safety critical and quality critical environments making black-box AI feasible into real industrial deployment by enhancing failure analysis.
- Produced interpretable visual outputs that enabled engineers and non-AI stakeholders to validate model behavior, supporting faster root-cause analysis and safer adoption of automated inspection systems.

Master's Thesis: A Retrieval-Augmented Framework for Fine-tuned LLMs for Vulnerability Analysis

- Developed a multi-phase framework to adapt large language models (LLaMA) for cybersecurity tasks through instruction tuning, domain-specific fine-tuning and retrieval-augmented generation.
- Curated and processed cybersecurity instruction datasets and Common Vulnerabilities and Exposures (CVE) data to enable progressive domain adaptation.
- Implemented and evaluated multiple fine-tuning strategies (instruction tuning, LoRA/Q-LoRA) and integrated FAISS-based retrieval for external knowledge access at inference time.
- Conducted a systematic quantitative evaluation comparing RAG-enabled and non-RAG models using established NLP metrics and statistical significance testing.
- Analyzed strengths, limitations and failure modes of domain-adapted LLMs in cybersecurity reasoning domain.

SKILLS

Python (Pandas, NumPy), SQL, ETL Workflows, Git, Docker, SCADA integration, ERP (JD Edwards), Linux/Unix basics, LSTM, RAG, Computer Vision.

LANGUAGES

English
IELTS: 8/9

German
B1-(actively improving)

SOCIAL NETWORKS

@gitesh-kumar
Gitesh,Kumar